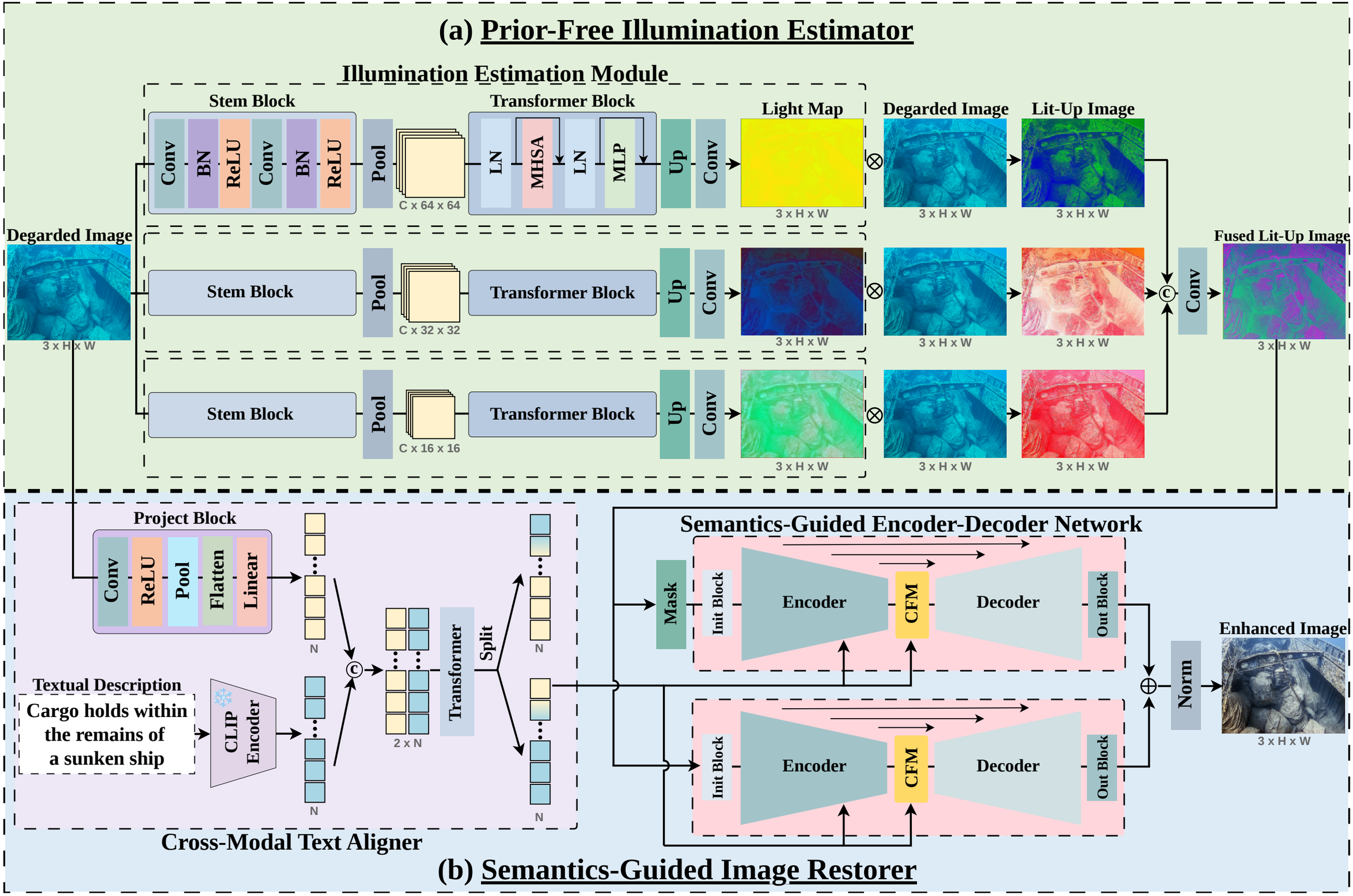
(a) Prior-Free Illumination Estimator
Illumination Estimation Module
Stem Block
Conv
BN
ReLU
Conv
BN
ReLU
Pool
C x 64 x 64
Transformer Block
LN
MHSA
LN
MLP
Up
Conv
Light Map
3 x H x W
Degarded Image
3 x H x W
Lit-Up Image
3 x H x W
Degarded Image
3 x H x W
Stem Block
Pool
C x 32 x 32
Transformer Block
Up
Conv
3 x H x W
3 x H x W
3 x H x W
Stem Block
Pool
C x 16 x 16
Transformer Block
Up
Conv
3 x H x W
3 x H x W
3 x H x W
Conv
Fused Lit-Up Image
3 x H x W
Project Block
Conv
ReLU
Pool
Flatten
Linear
N
Textual Description
Cargo holds within the remains of a sunken ship
CLIP Encoder
N
2 x N
Transformer
Split
N
N
Cross-Modal Text Aligner
Semantics-Guided Encoder-Decoder Network
Mask
Init Block
Encoder
CFM
Decoder
Out Block
Init Block
Encoder
CFM
Decoder
Out Block
Norm
Enhanced Image
3 x H x W
(b) Semantics-Guided Image Restorer